

Akshay Bambore

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Education

- PhD, Dept of Aero-Thermo-Mechanics- University Libre de Bruxelles, Belgium 2024-Present
- Masters in Aerospace Engineering, Sapienza University of Rome, Italy 2017-2020
- Bachelors of Mechanical Engineering, VTU Belgaum India 2012-2016

Research Experience

Dept of Aero-Thermo-Mechanics

2024-Present

PhD Researcher

Project : Study, Modelling and Computing Pressure Losses in Hydrogen Pipeline

- Determine pressure losses (total pressure losses) in pipelines.
- Evolution over operating time (in terms of years/decades of use) following degradation of the internal surface condition of the pipeline due to hydrogen embrittlement (H2) under pressure and erosion.
- CFD Computing and Mathematical calculations
- 2D and 3D calculations of pipe flow and pressure losses
- Development of a wall erosion model
- Analysis of external experimental data, development of the GH2 pipe pressure loss test bench and validation of models based on experimental measurements

Funding

This research is part of the Win4Excellence project, which is supported by the SPW Economy Employment Research of the Walloon Region and the Walloon Region Recovery Plan, within the framework of the agreement no. 2310142.

Dept of Aero-Thermo-Mechanics

2019-2019

Research Intern

Project : Experimental and Characterization of Rainfall Injectors for Icing Wind Tunnel

- **Experimental Characterization & Testing:** Executed high-precision laser diffraction tests using the **SMPATEC HELOS-VARIO/KR** system to characterize the droplet size distribution of ten different rainfall nozzles.
- **Systems Operation & Integration:** Managed the setup and operation of a **Model Delta Autojet 2008+** spray control panel and **PulsaJet** nozzles to simulate various climatic conditions, including light, moderate, and heavy rainfall as well as fog.

Industry Experience

Tata Consultancy Services

2021-2024

Manual Software Tester

Roles and Responsibilities:

- Testing Validation Rules, Work Flow Rules and Approval Processes for new Tasks, field updates and Email alerts according to application requirements.
 - Creating and customizing new Users and testing the implementation of security and sharing rules for different users.
 - Generating Reports based on the requirement of the business users and creating Dashboards for various profiles.
 - Completed unit and regression tests on application and individual modules.
 - Used exceptional communication and interpersonal skills to engage key stakeholders in technical testing outcomes.
 - Helping dev team in running debugging logs for identifying the errors.
 - Maintained and customized Salesforce.com scopes such as users, roles, profiles, groups, record types,
 - Responsible for preparation of Test cycle/progress reports, analysis of defect metrics & implement improvement actions and Tracking, Defect Reporting activities and Risk Analysis using Azure Devops Behaviour Driven Development (BDD) as much as possible and written test cases.
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Language Skill

English – Proficient

French – Beginner

Publications

Akshay Bambore, Patrick Hendrick and Jean Philippe Ponthot, *Study, Modelling and Computing of Pressure Losses in GH₂ Pipelines*, Selected Papers from the SDEWES 2025 Conference on Sustainable Development of Energy, Water, and Environment Systems, MDPI Energies **2026**, 19(4), 885: <https://doi.org/10.3390/en19040885>

Akshay Bambore, Patrick Hendrick, Jean Philippe Ponthot, *Proceedings of the 20th Conference on Sustainable Development of Energy, Water, and Environment Systems*, SDEWES2025.0946, 1-9

Community Engagement

Member of The Brussels Institute for Thermal-fluid systems and clean Energy (BRITE) research group <https://brite-research.be/akshay-bambore>

Presentations

Oral

- Oral presentation at 20th Conference on Sustainable Development of Energy, Water, and Environment Systems SDEWES **2025** conference
- Oral presentation at PhD day at ULB-Brussels **2025**
- Oral presentation at PhD day at ULB-Brussels **2025**

Poster

Study, Modelling and Computing of Pressure Losses in GH₂ Pipelines; Akshay Bambore Patrick Hendrick and Jean Philippe Ponthot 2025, ULB- Brussels

Study, Modelling and Computing of Pressure Losses in GH₂ Pipelines; Akshay Bambore Patrick Hendrick and Jean Philippe Ponthot 2025, Grenoble, France

PhD Activities

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| • Grenoble Energy Conversion & Storage (ENGINE2025) | 24-28 th February 2025 |
| • SDEWES 2025 conference | 5-10 th October 2025 |
| • Journal Publication in Energies MDPI | 8 th February 2026 |
| • University Certificate in Hydrogen Applications and Technologies | 16 th Jan – 11 th Sept 2026 |
| • Managing your research project: planning, goals, and time | 29 th Jan 2026 |
| • Hydrogen Hiveminds Hacathon | 18-20 th March 2026 |
| • PhD and Career development training | 29 th April 2026 |
| • World Hydrogen Awards | 19 th May 2026 |
| • EpHyc Conference | 1-3 rd June 2026 |

I hereby declare that all the information furnished in this document is free of errors to the best of my knowledge.

Akshay Bambore